AlphaWind Park – Asset Performance Investigation

Stakeholder Brief | 1–7 March 2026

Scope: Six wind turbines | Seven-day operational investigation

1. Executive Summary

AlphaWind generated **642.3 MWh** during the investigation period, compared with estimated potential production of **727.0 MWh**. This represents a performance ratio of **88.4%** and a theoretical production shortfall of **84.7 MWh**.

Approximately **35.8 MWh** of this shortfall is associated with extreme wind conditions and turbine cut-out above 25 m/s. This is not an operationally recoverable loss. Excluding this component, the estimated operational shortfall is **48.9 MWh**.

Grid limitations were the largest actionable recorded cause, accounting for 27.0 MWh. Technical faults and scheduled maintenance contributed another 10.6 MWh. The remaining unexplained deviations totalled 11.4 MWh.

6 March accounted for approximately 62% of the total estimated shortfall, making it the most important date for operational review.

T04 was the main turbine-level concern, consistently producing approximately 90% of peer output during normal operation.

2. Park-Level Energy Loss Breakdown

Cause	Estimated loss (kWh)	Share
Extreme wind (WND)	35,786	42.3%
Grid limitation (GRD)	26,952	31.8%
Technical faults (FLT)	6,628	7.8%
Scheduled maintenance (MTN)	3,931	4.6%
Unexplained	11,360	13.4%
Total	**84,657**	**100%**

The WND estimate assumes potential production of 248.5 kWh per 10-minute interval above the 25 m/s cut-out threshold. This represents theoretical energy opportunity rather than

avoidable operational loss. Using zero expected production above cut-out reduces the estimated shortfall to **48.9 MWh**, with all other categories unchanged.

Grid limitations reduced park output to approximately 25% of expected production during recorded events. High-wind shutdowns occurred on 6 March and 7 March. An additional 10-minute shutdown on 7 March was identified from production and weather data despite being absent from the original event log.

3. Turbine-Level Findings

T04 is the highest-priority asset for further investigation.

Its overall performance ratio was 79.7%, compared with 88.7–91.4% for the other turbines. During normal operation, T04 produced approximately **90.1% of peer-benchmarked output** across wind speeds from 5 to 25 m/s.

Its maximum recorded output was 225 kWh per 10-minute interval, compared with approximately 250 kWh for the other turbines. The consistent proportional gap suggests a possible controller derating or metering/signal-scaling issue.

T04 also had the longest time since service (120 days) and 7.5 hours of recorded fault and maintenance stops. These observations justify inspection but do not establish the cause of underperformance.

Three additional unlogged derating periods were identified:

- T02: 3 hours on 2 March.
- T05: 2 hours on 6 March.
- T03: 1 hour on 4 March (lower confidence).

Their combined net estimated contribution was approximately 1.3 MWh.

4. Data Quality and Methodology

The original stop log contained a duplicate record, four missing event end timestamps and incomplete reporting of park-level events. The analysis removed duplicates, reconstructed missing timestamps where matching records were available, corrected an extended fault duration and added two missing grid-event records supported by production data.

Overlapping events were handled using the priority WND > FLT > MTN > GRD, ensuring that each turbine and 10-minute interval was attributed only once.

Expected production was estimated using a peer-based production curve constructed from five reference turbines, excluding T04. Event-affected observations were removed, and mean production was calculated in 0.5 m/s wind-speed intervals.

The baseline is based on only seven days of observations. It is an empirical benchmark, not a manufacturer-certified power curve. Results should therefore be interpreted as indicative estimates rather than precise measurements.

5. Recommended Actions

1. **Investigate T04:** Compare SCADA output with an independent meter, verify controller settings and signal scaling, and review its hydraulic fault and service history.
2. **Verify grid limitations:** Reconcile export restrictions on 4 and 6 March with control-room and grid-operator records.
3. **Investigate unlogged derating:** Review alarm and controller records for T02, T05 and T03.
4. **Improve event reporting:** Ensure park-level events are consistently recorded across affected turbines, with accurate start and end timestamps.
5. **Improve maintenance scheduling:** Where operationally feasible, schedule flexible maintenance during lower-wind periods.

Further Work (TODO)

Validate the reference curve against longer historical data or the manufacturer's power curve. Test sensitivity to reference-turbine selection and wind-speed binning. Incorporate turbine-level wind measurements and quantify uncertainty around the loss estimates.

Conclusion

The main priorities are investigating T04's persistent output gap and verifying grid-curtailment records. High-wind cut-out explains a substantial theoretical production opportunity but should not be considered an avoidable operational loss.

The findings provide a basis for targeted operational follow-up, with further validation against SCADA, independent meter and grid-dispatch records recommended before formal performance reporting.

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